

PYONGGANG, Korea, Democratic People's Republic of

WMO: 470750

Lat: 38.417N

Lon: 127.283E

Elev: 352

StdP: 97.17

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.1	-15.8	-22.2	0.5	-15.9	-20.2	0.6	-13.8	8.2	-8.5	7.4	-6.7	1.3	230	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.4	30.1	24.4	28.7	23.3	27.4	22.5	25.5	28.8	24.7	27.4	23.9	26.1	2.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.5	20.3	27.5	23.8	19.5	26.3	23.2	18.8	25.5	80.8	29.0	77.2	27.6	74.2	26.2	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.9	6.0	-21.0	32.5	2.9	1.6	-23.0	33.6	-24.7	34.6	-26.3	35.4	-28.4	36.6		
(5)			-21.0	26.7	2.9	0.9	-23.0	27.4	-24.7	28.0	-26.3	28.5	-28.4	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-6.7	-3.6	2.5	9.1	15.5	20.1	23.0	23.4	18.0	11.7	3.8	-4.5		
	DBStd	11.04	4.52	4.21	3.71	3.83	2.82	2.12	2.25	2.44	2.67	3.38	4.62	4.71		
	HDD10.0	1856	519	381	233	62	1	0	0	0	0	22	190	448		
	HDD18.3	3644	777	614	489	277	93	8	1	1	37	204	435	707		
	CDD10.0	1649	0	0	2	34	172	303	402	414	241	76	5	0		
	CDD18.3	397	0	0	0	0	6	61	144	158	28	1	0	0		
	CDH23.3	2289	0	0	0	4	104	356	732	971	118	4	0	0		
	CDH26.7	466	0	0	0	0	9	56	158	239	3	0	0	0		
(14) Wind	WSAvg	2.0	1.8	2.0	2.5	2.7	2.5	2.1	2.0	1.5	1.5	1.7	2.1	2.1		
(15) Precipitation	PrecAvg	1416	17	26	30	67	103	140	447	328	136	54	47	25		
	PrecMax	2259	49	98	77	160	324	401	1166	920	424	222	138	80		
	PrecMin	682	1	0	6	11	14	52	154	43	8	4	10	5		
	PrecStd	329	12	22	19	36	63	89	230	211	103	48	29	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	10.3	17.0	23.9	27.8	30.0	32.2	32.4	27.1	23.5	17.2	7.8		
		MCWB	2.8	6.6	9.8	13.6	17.2	21.2	25.4	26.2	20.4	16.7	12.4	5.9		
	2%	DB	2.9	6.8	13.8	21.1	25.8	28.2	29.7	30.6	25.9	21.4	14.4	5.7		
		MCWB	0.7	2.5	8.6	12.0	16.3	20.4	24.5	25.2	20.2	15.4	10.7	3.7		
	5%	DB	1.5	4.7	11.3	18.9	24.3	26.8	28.2	29.2	24.8	19.6	12.4	3.9		
		MCWB	-0.1	1.5	7.1	11.1	15.7	20.0	23.7	24.5	19.4	14.3	9.1	2.1		
	10%	DB	0.3	2.8	9.0	16.8	22.3	25.4	26.9	27.8	23.6	18.1	10.8	2.3		
		MCWB	-1.2	0.3	5.2	10.2	15.1	19.2	23.2	23.7	18.8	13.4	8.3	0.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	8.1	11.6	15.6	19.4	22.9	26.3	26.9	23.2	18.3	13.5	6.3		
		MCDB	4.7	10.6	14.5	21.8	24.2	27.3	30.1	31.5	25.3	21.7	15.9	7.4		
	2%	WB	1.4	3.5	9.3	13.5	18.0	21.8	25.2	25.8	21.7	16.4	11.3	4.0		
		MCDB	2.4	5.7	13.0	18.3	22.6	26.0	28.5	29.4	24.0	19.6	13.4	5.4		
	5%	WB	0.1	1.8	7.4	12.2	16.9	21.0	24.4	25.1	20.5	15.2	9.9	2.3		
		MCDB	1.3	4.1	10.9	17.2	22.0	24.9	27.1	28.2	23.3	18.6	12.0	3.6		
	10%	WB	-1.1	0.4	5.5	10.9	16.0	20.2	23.8	24.4	19.6	14.2	8.5	1.0		
		MCDB	0.3	2.5	8.6	15.8	20.8	23.8	25.9	26.8	22.5	17.3	10.4	2.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	9.9	9.6	11.8	11.4	9.4	6.3	7.4	10.1	11.5	9.1	8.8		
		MCDBR	9.8	11.5	13.3	15.1	15.1	12.3	8.9	9.4	11.7	13.3	11.2	8.9		
	5% WB	MCWBR	7.9	7.9	8.2	7.5	6.9	5.7	4.4	4.7	6.1	7.6	7.4	7.1		
		MCWBR	8.5	9.6	12.5	13.1	11.9	9.9	7.0	8.1	8.9	11.6	9.3	8.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.391	0.454	0.533	0.592	0.597	0.662	0.634	0.543	0.474	0.475	0.446	0.391		
		taud	1.966	1.850	1.743	1.671	1.700	1.626	1.747	1.971	2.080	1.954	1.959	2.011		
	Ebn at Noon	Ebn at Noon	736	737	720	709	717	670	686	741	770	710	670	702		
		Edh at Noon	133	171	211	240	237	256	225	176	149	152	134	119		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.43	3.27	4.16	4.89	5.48	5.12	3.92	4.15	4.05	3.49	2.37	2.17		
	RadStd	RadStd	0.18	0.22	0.41	0.41	0.54	0.48	0.58	0.55	0.46	0.25	0.27	0.19		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page